

Full Length Article

DFIR-DETR: Frequency-domain iterative refinement and dynamic feature aggregation for small object detection

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ARTICLE INFO

Keywords:

Small object detection
Transformer-based detection
Frequency-domain feature learning
Sparse attention mechanism
Multi-scale feature fusion
Cross-scene generalisation

ABSTRACT

Small object detection in complex scenes exposes a fundamental tension in neural network design: backbone attention distributes computation uniformly regardless of content, pyramid necks inflate activation magnitudes during upsampling without norm compensation, and bottleneck convolutions progressively smooth high-frequency edge components through accumulated spatial filtering. In response, we develop DFIR-DETR by tracing each proposed module back to a specific, measurable deficiency in the RT-DETR baseline: uniform attention that ignores spatial complexity, norm drift that destabilises upsampled features, and spatial convolutions that progressively suppress the high-frequency components small objects depend on. On NEU-DET and VisDrone, DFIR-DETR achieves 92.9% and 51.6% mAP50 with only 11.7M parameters and 47.2 GFLOPs, demonstrating consistent gains across two qualitatively different detection domains.

1. Introduction

Small object detection remains one of the most demanding problems in deep neural network research. Targets occupying fewer than 32×32 pixels carry limited texture information, forcing the network to rely heavily on boundary signals and long-range contextual cues that standard architectures are poorly equipped to exploit. The practical stakes differ sharply between the two application domains this work targets. On industrial production lines, a micro-defect missed during hot-rolled steel inspection does not stay missed; mechanical stress concentrates at the undetected boundary and the failure propagates downstream. Throughput constraints are equally unforgiving: frame budgets measured in tens of milliseconds rule out the multi-pass strategies that offline detectors rely on for accuracy. Aerial monitoring from UAVs presents a structurally different version of the same difficulty. At cruising altitude, a pedestrian or cyclist collapses to 10–15 pixels; the surrounding scene is dense and cluttered, so both missed targets and spurious detections carry real operational cost. What unites the two domains is a shared bottleneck: neither can tolerate the loss of high-frequency boundary information that deep spatial filtering progressively introduces. What separates them is everything else: background statistics, object formation mechanism, scale distribution, imaging geometry. A detector that holds up across both is harder to dismiss as a dataset-specific artefact.

The difficulty is further compounded in cross-scene settings, where a single model must generalise across domains as different as UAV aerial imagery and industrial surface inspection.

A closer examination of modern real-time detectors reveals three structural limitations that bear directly on small object performance. First, convolutional backbones allocate attention computation uniformly across the spatial domain, giving equal weight to uninformative backgrounds and information-dense object boundaries (Yuan et al., 2024). Second, feature pyramid necks expand activation magnitudes during upsampling without compensating normalisation, disrupting gradient dynamics and degrading cross-scale feature fusion (Wang & Deng, 2024). Third, repeated spatial convolutions act as implicit low-pass filters, attenuating the high-frequency edge components that small objects depend upon for accurate localisation (Chi et al., 2020).

RT-DETR (Zhao et al., 2024) represents the current state of the art in real-time transformer detection, combining a CNN backbone with a hybrid encoder and an end-to-end query based decoder. However, its ResNet backbone applies fixed-budget attention without content adaptation, its CCFF neck upsamples without amplitude control, and its RepC3 bottleneck operates entirely in the spatial domain, leaving all three limitations unaddressed.

We propose DFIR-DETR, which introduces three architectural contributions corresponding to each identified limitation. The Dynamic Content-Feature Aggregation (DCFA) module parsifies the attention

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matrix through a dynamic Top-K mechanism predicted from local feature statistics, reducing complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(NK)$. The Dynamic Feature Pyramid Network (DFPN) introduces norm-preserving upsampling together with a dual-path convolution structure for explicit spatial detail recovery. The Frequency domain Iterative Refinement module (FIRC3) reformulates feature aggregation as a constrained optimisation problem in the spectral domain, where high-frequency boundary components can be explicitly reinforced.

The main contributions of this work are summarised as follows:

- DCFA is introduced as a content-adaptive backbone module, where a dynamic Top-K sparsification mechanism redistributes attention capacity toward structurally complex regions, bringing complexity down from $\mathcal{O}(N^2)$ to $\mathcal{O}(NK)$ without sacrificing global context modelling.
- Drawing on an L_1 -norm conservation argument, DFPN is developed with analytically derived amplitude normalisation during upsampling and a dual-path shuffle convolution for explicit fine-grained spatial detail recovery across pyramid scales.
- The bottleneck feature aggregation problem is reformulated as a frequency-domain least-squares optimisation in FIRC3, granting the network direct and learnable access to high-frequency boundary components that accumulate spatial filtering cannot preserve.
- On NEU-DET and VisDrone, the proposed architecture attains 92.9% and 51.6% mAP50 with only 11.7M parameters and 47.2 GFLOPs, achieving state-of-the-art accuracy while reducing both model size and computational cost relative to the baseline.

The remainder of this paper is organised as follows. Section 2 reviews related work on object detection architectures, attention mechanisms, multi-scale feature fusion, and frequency-domain neural network modules. Section 3 presents the proposed DFIR-DETR framework and details the design of DCFA, DFPN, and FIRC3. Section 4 describes the experimental setup, benchmark datasets, ablation studies, and comparisons with state-of-the-art methods. Section 5 concludes the paper.

2. Related work

2.1. Object detection architectures

Object detection has undergone two broad paradigm shifts over the past decade. Two-stage methods such as Faster R-CNN (Ren et al., 2017a) separated region proposal from classification, achieving strong accuracy at the cost of inference speed. One-stage detectors including SSD (Liu et al., 2016) and YOLO (Redmon et al., 2016) collapsed this pipeline into a single forward pass, with YOLOv3 (Redmon & Farhadi, 2018) demonstrating that multi-scale prediction heads could recover much of the accuracy lost by removing the proposal stage.

DETR (Carion et al., 2020) subsequently recast detection as set prediction via bipartite matching, discarding both anchors and non-maximum suppression. Deformable DETR (Zhu et al., 2021), DN-DETR (Li et al., 2022), and DINO (Zhang et al., 2023a) progressively addressed its convergence and accuracy limitations. RT-DETR brought this line of work to practical deployment by pairing a hybrid encoder with a query-based decoder, achieving real-time throughput while retaining end-to-end training. The present work takes RT-DETR as its starting point, leaving the decoder unchanged while substantially rethinking how features are extracted, fused, and refined prior to decoding.

2.2. Attention mechanisms and sparse computation

Attention has become the central computational primitive in modern vision networks, yet its quadratic cost with respect to sequence length remains a persistent bottleneck. Local window attention (Liu et al., 2024) confines each token to a bounded spatial neighbourhood, improving throughput but weakening the long-range dependency modelling critical for small object detection. Deformable attention (Zhu et al., 2021)

maintains linear cost by sampling a fixed number of reference points per query, but loses the ability to adapt to local feature complexity. Token merging approaches (Bolya et al., 2023) combine redundant tokens to reduce sequence length, yet the resulting loss of positional specificity is detrimental for localisation-sensitive tasks. In detection, sparse query designs appear in Sparse DETR (Roh et al., 2022) and Efficient DETR (Yao et al., 2021), which reduce computational load by selecting informative encoder tokens. However, these methods fix the sparsification threshold globally, whereas the informational density of a feature map varies substantially between defect-heavy industrial images and sparsely populated aerial frames. DCFA addresses this by predicting the retention ratio from local feature statistics, making sparsification content-dependent rather than scene-agnostic, and pairs the resulting sparse attention with a spatial gated linear unit that enriches nonlinear channel transformations with neighbourhood context.

2.3. Multi-scale feature fusion

Feature pyramids (Lin et al., 2017a) established the principle of combining semantic richness from deep layers with spatial precision from shallow layers through a top-down pathway with lateral connections. PANet (Liu et al., 2018) augmented this with a complementary bottom-up path, shortening the information routing distance between low-level detail and high-level semantics. BiFPN, introduced within EfficientDet (Tan et al., 2020), went further by learning scalar weights for each cross-scale connection, allowing the network to modulate the relative contribution of each resolution level during fusion. Despite these refinements, a complication common to all pyramid designs has received little attention: upsampling expands feature map area by s^2 , inflating the L_1 norm of activations and disrupting gradient balance between upsampled and lateral paths. SlimNeck (Li et al., 2024) and Gold-YOLO (Wang et al., 2023a) improve feature routing efficiency but leave this amplitude drift uncorrected. DFPN compensates through an analytically derived normalisation factor, providing stable cross-scale fusion that prior designs do not offer.

2.4. Frequency-domain representations in neural networks

The motivation for incorporating spectral operations into detection networks has both efficiency and representational grounds. Phase-aware MLP architectures (Tang et al., 2022) demonstrated that decomposing image patches into amplitude and phase components enables richer spatial modelling than purely spatial mixing, establishing that frequency-domain parameterisation captures structure that spatial convolutions miss. Beyond representational benefits, the computational argument is compelling: a convolution of kernel size k over a spatial map of size N costs $\mathcal{O}(Nk^2)$, whereas the equivalent spectral operation requires only $\mathcal{O}(N \log N)$ regardless of effective receptive field size.

Among classification-oriented frequency methods, FcaNet (Qin et al., 2021) decomposes channel attention into discrete cosine components, GFNet (Rao et al., 2021) substitutes spatial token-mixing layers with global Fourier filters, and FocalNet (Yang et al., 2022b) aggregates context at multiple scales through focal modulation. These methods enrich feature representations within a single resolution stage but do not engage with the cross-scale fusion problem. Within the detection literature, SF-DETR (Wang & Gao, 2025) injects scale-frequency awareness into the encoder to capture high-frequency boundary cues in drone-view scenes, and Freq-DETR (Chen et al., 2025) constructs dual-branch processing to jointly model spatial and frequency features for UAV imagery. Both methods treat frequency as a supplementary signal alongside spatial features, however, rather than as the exclusive medium for bottleneck computation. FIRC3 departs from this pattern by fully relocating feature refinement into the spectral domain and solving the resulting problem as a constrained least-squares optimisation, giving the network direct and learnable control over the high-frequency content that governs boundary localisation for small objects.

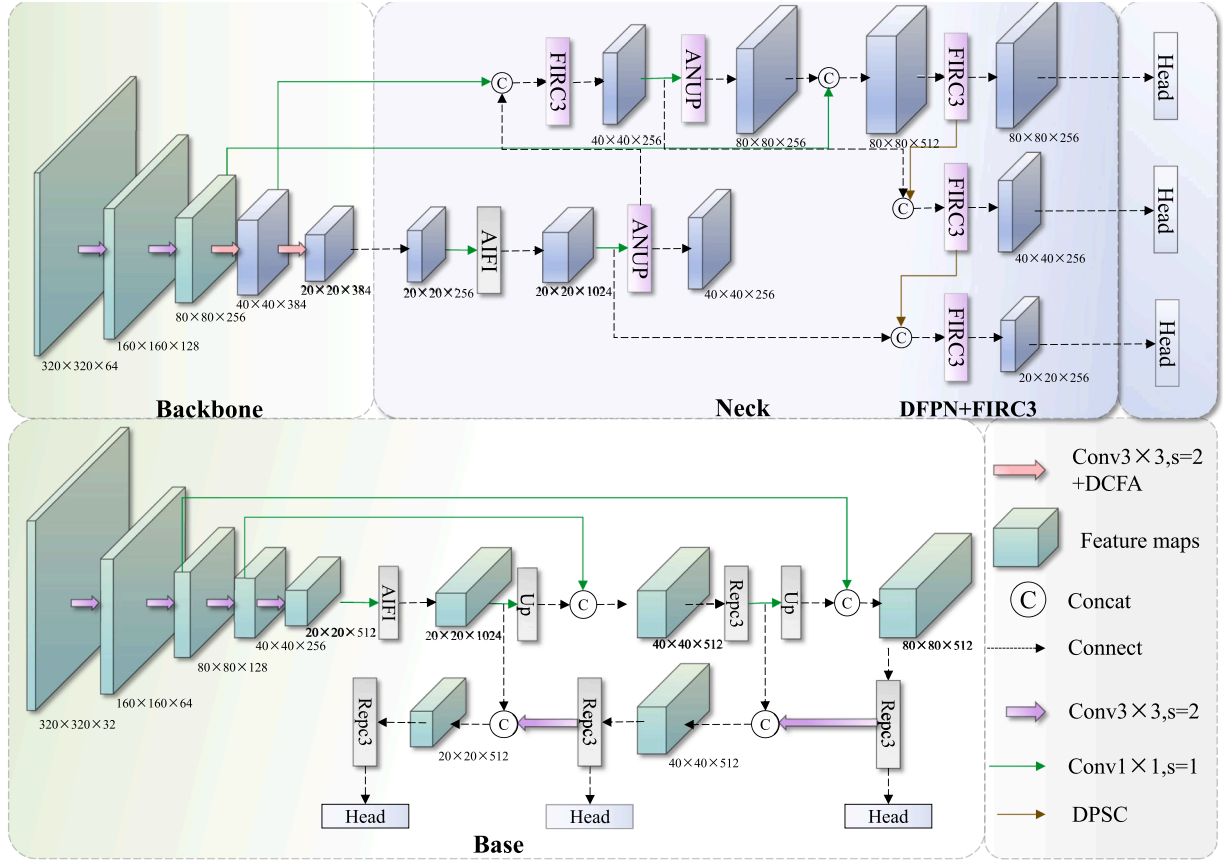


Fig. 1. Overall architecture of DFIR-DETR.

3. Methods

We propose DFIR-DETR to address small object detection challenges in UAV remote sensing and industrial surface inspection. The architecture, illustrated in Fig. 1, tackles two fundamental problems: sparse feature representation of distant small objects in VisDrone and subtle defect texture capture in NEU-DET. DCFA serves as the Backbone, where content-adaptive K-sparse attention allocates computational resources dynamically—defect regions and small targets receive concentrated modeling while uniform backgrounds undergo aggressive pruning. DFPN functions as the Neck, preventing feature inflation during upsampling through amplitude normalization and retaining fine spatial details through dual-path shuffle convolution across scales. FIRC3 operates in the feature fusion layer, processing features in the frequency domain to capture global context at reduced computational cost. Each component targets specific failure modes of existing detectors: DCFA addresses limited receptive fields, DFPN prevents information loss during scale transitions, and FIRC3 preserves high-frequency boundary details.

3.1. DCFA

The backbone of conventional RT-DETR relies on a ResNet architecture composed of homogeneous BasicBlock modules stacked uniformly across all stages. However, this design exhibits three limitations that become particularly pronounced in cross-scene small object detection. Progressive downsampling erodes the fine-grained geometric detail and edge continuity that small objects and subtle surface defects depend upon, while the fixed receptive field expansion schedule lacks the flexibility to accommodate the wide scale range encountered across different scenes. Furthermore, without explicit spatial attention, the network cannot suppress irrelevant background activations in dense or cluttered

environments, leaving both false positive and false negative rates persistently high. To address these limitations, we propose DCFA, a content-adaptive feature aggregation backbone that treats representational allocation as a learnable function of local feature complexity. The key innovation is learning to allocate computational resources based on local complexity: small object regions receive concentrated attention modeling through dynamic K-sparse selection (DKSA), while uniform backgrounds are efficiently pruned. Spatial gated linear units (SGLU) further incorporate neighborhood context into nonlinear transformations. This design simultaneously preserves fine-grained geometric details and high-level semantic information, as illustrated in Fig. 2a.

The DCFA module adopts a cross-stage partial (CSP) network architecture, achieving gradient-efficient propagation and multi-scale feature fusion through dual-path feature flows. The complete feature transformation process can be mathematically formulated as:

$$Y = \phi_{cv2} \left(\text{Concat} [F_1, F_2, G_1(F_2), G_2(G_1(F_2)), \dots, G_n(G_{n-1}(\dots))] \right) \quad (1)$$

$$[F_1, F_2] = \text{split}(\phi_{cv1}(X)) \quad (2)$$

where G_i represents the i th DAFB (Dynamic Attention Fusion Block) transformation operator, n denotes the stacking depth, and ϕ_{cv1} and ϕ_{cv2} represent the input and output 1×1 convolutional projection layers, respectively.

DAFB achieves comprehensive feature enhancement by cascading three functionally complementary sub-modules. Its forward propagation follows the transformation chain:

$$Z = \text{SGLU}(H + \text{DKSA}(H)), \quad H = X + \phi_{dw}(X) \quad (3)$$

where ϕ_{dw} denotes a 3×3 depthwise separable convolution with batch normalization. The DKSA mechanism subsequently operates on the enhanced features H , selectively focusing on defect regions in industrial

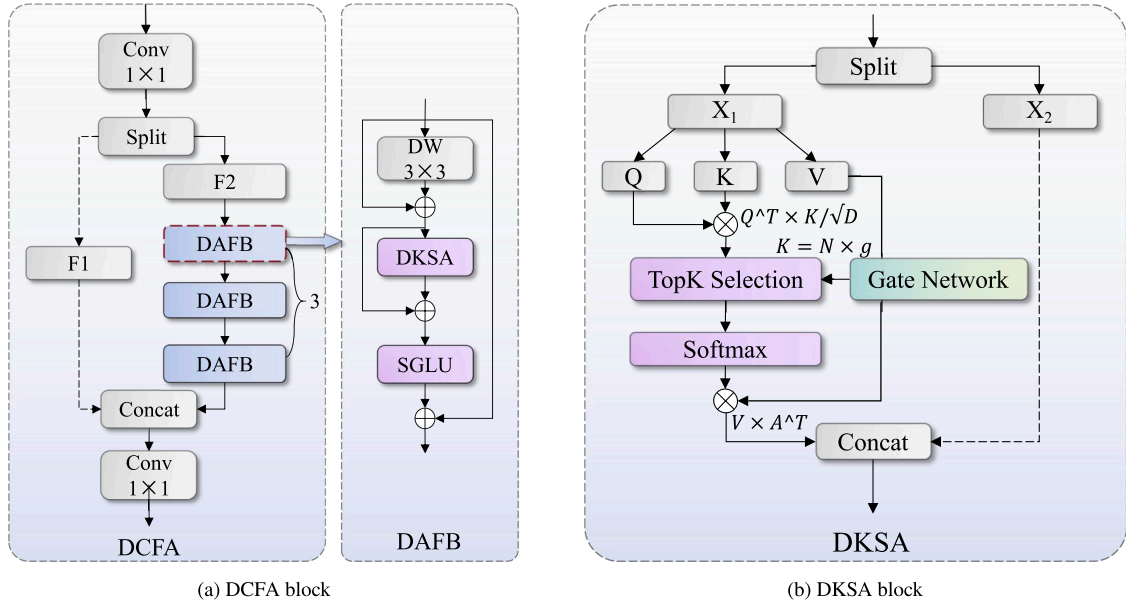


Fig. 2. (a) DCFA block; (b) DKSA block.

scenarios through dynamic sparsification strategies while establishing long-range associations between small objects and contexts in remote sensing scenarios. SGLU serves as a feedforward network to perform nonlinear transformation and channel mixing on attention-enhanced features. Its gating mechanism adaptively modulates the activation intensity of different channels, further improving cross-scene feature discriminability. The mathematical expression is:

$$\text{SGLU}(X) = X + \phi_{out} \left(D \left(\left(\phi_{act} \left(\phi_{dw}^{3 \times 3} (X_g) \right) \right) \odot X_v \right) \right) \quad (4)$$

$\phi_{dw}^{3 \times 3}$ represents the 3×3 depthwise convolution applied to the gating stream, encoding local spatial contextual information into gating signals so that gating decisions integrate neighborhood spatial patterns rather than relying solely on single-point feature values. ϕ_{act} employs the GELU activation function, providing smoother gradient flow compared to ReLU. Element-wise multiplication $\odot$ implements gating modulation, and D denotes Dropout regularization.

DKSA implements computationally efficient self-attention through a dynamic Top-K selection strategy, demonstrating excellent adaptability in both industrial defect detection and remote sensing small object recognition scenarios, as illustrated in Fig. 2b. Combined with LGN (Layer Group Normalization) preprocessing, the complete attention computation process can be uniformly expressed as:

$$\text{DKSA}(X) = \phi_{proj} \left(\text{Concat} \left[(V A^T)^{reshape}, X_2 \right] \right) \quad (5)$$

$$A_{ij} = \begin{cases} \frac{\exp(s_{ij})}{\sum_{j' \in \mathcal{T}_K^i} \exp(s_{ij'})}, & j \in \mathcal{T}_K^i \\ 0, & j \notin \mathcal{T}_K^i \end{cases} \quad (6)$$

The dynamic Top-K selection mechanism is defined as:

$$K = \lfloor N \cdot \sigma(\text{AvgPool}(\psi(X))) \rfloor \quad (7)$$

where ψ represents a gating network composed of two convolutional layers, σ denotes the sigmoid function, and the output scalar value determines the proportion of attention connections to retain. $\mathcal{T}_K^i = \text{TopK}(S_i, K)$ returns the index set of the top K most relevant key positions for the i th query position. Unselected connections are masked to zero, thereby converting the attention matrix from dense to structurally

sparse, reducing computational complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(NK)$ (Algorithms 1–3).

Algorithm 1 DCFA.

Require: Input feature map $X \in \mathbb{R}^{C \times H \times W}$, stack depth N

Ensure: Enhanced feature map $Y \in \mathbb{R}^{C \times H \times W}$

```

1:  $X_1, X_2 \leftarrow \text{Split}(X)$  // Dual-path feature flow
2:  $F \leftarrow X_1$ 
3: for  $i = 1$  to  $N$  do
4:   // Dynamic Attention Fusion Block
5:    $F' \leftarrow \text{DWConv}_{3 \times 3}(F) + \text{BN}(F)$ 
6:   // Dynamic K-Sparse Attention
7:    $F'' \leftarrow \text{LGN}(F')$ 
8:    $Q, K, V \leftarrow \text{Linear}(F'')$ 
9:    $\rho \leftarrow \sigma(\text{Gate}(F''))$ 
10:   $k \leftarrow \lfloor \rho \cdot HW \rfloor$ 
11:  for each position  $j$  in  $[1, HW]$  do
12:     $\mathcal{T}_K^j \leftarrow \text{TopK}(Q_j \cdot K^T, k)$ 
13:     $\text{Attn}_j \leftarrow \text{Softmax} \left( \frac{Q_j \cdot K^T}{\sqrt{d}} \right) \cdot V_{\mathcal{T}_K^j}$ 
14:  end for
15:  // Spatial Gated Linear Unit
16:   $G \leftarrow \text{GELU}(\text{DWConv}_{3 \times 3}(\text{Attn}))$ 
17:   $F \leftarrow \text{Dropout}(\text{Linear}(\text{Attn}) \odot G)$ 
18: end for
19:  $Y \leftarrow \text{Conv}_{1 \times 1}(\text{Concat}(F, X_2))$ 
20: return  $Y$ 

```

The dynamic K-sparse attention pruning mechanism enables content-adaptive allocation of computational resources, focusing more attention connections on defect regions in industrial scenarios to capture subtle textures while performing aggressive pruning on uniform backgrounds to reduce redundancy. In remote sensing scenarios, it allocates sufficient contextual modeling capability to small objects while sparsifying large irrelevant backgrounds, achieving optimal trade-offs between accuracy and efficiency in both scenarios. The spatial gated linear unit provides nonlinear modeling capability far exceeding traditional ReLU through spatialized gating signals. Its smooth gradient characteristics avoid gradient vanishing problems, enabling stable training of deep networks

Algorithm 2 DFPN.**Require:** Multi-scale features $\{F_i\}_{i=1}^L$, scale s **Ensure:** Enhanced features $\{F_i^{\text{out}}\}_{i=1}^L$

```

1:  $F_L^{\text{up}} \leftarrow F_L$ 
2: // Top-down with ANUP
3: for  $i = L - 1$  down to 1 do
4:    $F_i^{\text{up}} \leftarrow \text{ANUP}(F_{i+1}^{\text{up}}, F_i, s)$ 
5: end for
6: // Apply DPSC
7: for  $i = 1$  to  $L$  do
8:    $F_i^{\text{out}} \leftarrow \text{DPSC}(F_i^{\text{up}})$ 
9: end for

10: Procedure ANUP( $F_{\text{high}}, F_{\text{low}}, s$ ):
11:    $F_{\text{up}} \leftarrow \text{Upsample}(F_{\text{high}}, s)$ 
12:    $F_{\text{norm}} \leftarrow (1/s^2) \cdot F_{\text{up}}$  // Amplitude normalization
13:   return Conv $_{1 \times 1}$ (Concat( $F_{\text{norm}}, F_{\text{low}}$ ))

14: Procedure DPSC( $F$ ):
15:    $F_1 \leftarrow \sigma(W_{\text{std}} * F)$  // Path 1
16:    $F_2 \leftarrow \text{GELU}(W_d * \text{GELU}(W_{\text{pw}} * F))$  // Path 2
17:   return ChannelShuffle(Concat( $F_1, F_2$ ))

```

and learning of more refined feature transformations. The learnable gating mechanism in Eq. (7) adaptively determines K based on local feature statistics, which can be interpreted as adaptively balancing representational capacity against computational budget. This enables the network to automatically adjust the trade-off between efficiency and accuracy for each input sample based on scene complexity.

3.2. DFPN

The cross-scale feature fusion module CCFF in conventional RT-DETR relies on simple nearest neighbor upsampling and direct concatenation operations for multi-scale feature integration. However, abrupt transitions between feature scales introduce semantic discontinuities, compromising feature representation quality with particularly significant impact on small objects where spatial resolution is critical. The symmetric top-down pathway lacks sufficient consideration of information density differences between scales, resulting in suboptimal feature fusion performance in small object detection tasks. To address these fundamental deficiencies of CCFF, we propose DFPN, a principled redesign of multi-scale feature fusion guided by two core insights: upsampling must preserve feature intensity to prevent gradient instability, and downsampling must retain spatial details critical for small object boundaries. We achieve this through amplitude-normalized upsampling (ANUP) in the top-down pathway that maintains consistent feature magnitudes across scales, and dual-path shuffle convolution (DPSC) in the bottom-up pathway that explicitly preserves fine-grained spatial information during aggregation.

DFPN addresses feature inflation through amplitude-normalised upsampling in the top-down pathway and recovers spatial detail via dual-path shuffle convolution in the bottom-up pathway, as illustrated in Figs. 3 and 4.

In the top-down pathway, the ANUP module reconstructs traditional upsampling operations by introducing amplitude-dependent normalization factors to compensate for feature amplitude inflation caused by spatial expansion. Given an input feature map F_i at scale i , the ANUP operation produces amplitude-controlled upsampled feature maps:

$$F_i^{\uparrow} = \beta \cdot \mathcal{U}(F_i) = \frac{1}{s^2} \cdot \mathcal{U}(F_i) \quad (8)$$

where $\mathcal{U}(\cdot)$ denotes the nearest neighbor upsampling operator with scale factor s , and $\beta = 1/s^2$ represents the amplitude normalization coefficient, maintaining consistent feature intensity across different scales.

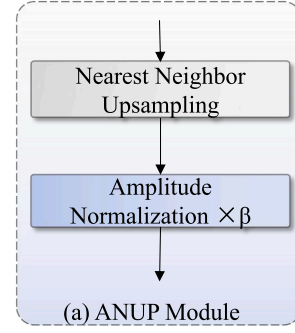


Fig. 3. ANUP module.

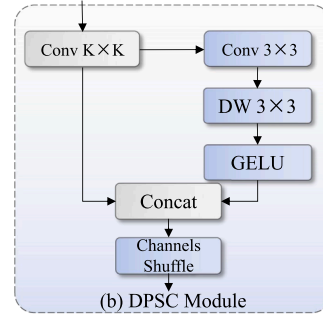


Fig. 4. DPSC module.

This normalization strategy ensures that upsampled features retain original information density, preventing the feature map inflation common in traditional CCFF upsampling operations. The theoretical foundation stems from the fact that spatial interpolation inherently amplifies the feature map L_1 norm by a factor proportional to the square of the up-sampling ratio. The proposed amplitude normalization counteracts this effect by maintaining the relationship:

$$\|F_i^{\uparrow}\|_1 \approx \|F_i\|_1 \cdot \frac{s^2 \cdot H \cdot W}{H \cdot W} \cdot \frac{1}{s^2} = \|F_i\|_1 \quad (9)$$

establishing a more stable feature propagation mechanism, which is particularly significant for preserving subtle appearance features of small objects.

In the bottom-up pathway, the DPSC module addresses information loss during downsampling through a dual-path architecture. The first path performs standard convolution to extract semantic features, while the second path takes the output of the first path as input, capturing spatial details through cascaded convolutions, ultimately fusing dual-path information through channel shuffling. The DPSC operation is formulated as:

$$F_{\text{out}} = \mathcal{M}(\text{Concat}(F_1^{\text{std}}, \phi(W_d * (W_{\text{conv}} * F_1^{\text{std}})))) \quad (10)$$

F_1^{std} represents semantic features extracted through standard convolution with activation function $\sigma(\cdot)$, W_{conv} and W_d denote the 3×3 standard convolution kernel and 3×3 depthwise convolution kernel (groups = $C/2$) in the second path respectively, $\phi(\cdot)$ represents the GELU activation function providing smooth gradient flow, and $\mathcal{M}(\cdot)$ denotes the channel shuffle operation. The channel shuffle mechanism is mathematically represented as a tensor rearrangement operation:

$$\mathcal{M}(F) = \text{Reshape}(\text{Permute}(\text{Reshape}(F, [B, 2, C/2, H, W]), [0, 2, 1, 3, 4]), [B, C, H, W]) \quad (11)$$

reorganizing feature channels to facilitate cross-path information exchange. The second path in DPSC constitutes a spatial detail capture

Table 1
Ablation study on NEU-DET dataset.

DCFA	DFPN	FIRC3	mAP50	Cr	In	Pa	Ps	Ri	Sr	mAP50:95	Params	GFLOPs	FPS
×	×	×	88.7	78.4	87.0	96.0	94.6	78.0	98.1	58.2	19.9	57.0	36.9
✓	×	×	90.1	78.9	91.2	97.0	96.8	84.0	98.6	60.8	13.7	51.7	37.0
×	✓	×	90.9	72.1	94.5	97.0	94.4	89.2	98.3	62.6	19.7	56.7	33.7
×	×	✓	90.5	76.2	94.6	95.8	92.8	84.7	98.6	63.2	18.2	48.4	47.8
✓	✓	×	92.4	86.1	92.4	96.6	98.8	84.8	95.6	64.0	13.4	49.8	40.2
✓	×	✓	91.7	82.9	92.1	94.8	98.3	83.2	99.1	62.1	14.2	48.2	37.6
×	✓	✓	91.1	81.3	89.9	96.7	95.1	84.7	98.7	61.6	15.1	47.3	33.7
✓	✓	✓	92.9	84.0	93.7	96.8	98.4	84.9	99.3	65.9	11.7	47.2	38.0

branch, taking the output F_1^{std} of the first path as input and achieving fine-grained feature extraction through cascaded convolution operations:

$$F_2^{dep} = \phi(W_d * (W_{conv} * F_1^{std})) \quad (12)$$

where W_{conv} is a 3×3 standard convolution kernel for initial spatial feature transformation, and W_d is a 3×3 depthwise convolution kernel for efficient spatial filtering. Through grouped operations, spatial features are progressively refined while maintaining computational efficiency, with GELU activation ensuring smooth gradient propagation required for training stability. The concatenation and shuffling of features from both paths establish comprehensive feature representations, integrating semantic information from the standard convolution path with fine-grained spatial details from the cascaded convolution path.

ANUP stabilises top-down propagation by eliminating the norm inflation inherent in standard upsampling, ensuring high-level semantic information reaches lower scales with preserved intensity, a property particularly valuable for small objects that depend heavily on contextual cues. DPSC complements this by routing fine-grained boundary detail through a dedicated cascaded convolution stream, so spatial precision is not sacrificed during bottom-up aggregation. The result is more consistent cross-scale feature flow and improved localisation accuracy for small objects across both industrial and aerial detection scenarios.

3.3. FIRC3

Conventional RT-DETR employs the RepC3 module for multi-scale feature fusion in its feature pyramid network and path aggregation network structures. However, RepC3 primarily relies on spatial domain convolution operations, exhibiting inherent limitations in capturing long-range dependencies and processing small object features. The spatial locality constraint of conventional convolutions limits the receptive field scope, resulting in insufficient feature representation capability for small objects, while contextual information is crucial for small object detection. Repeated application of spatial domain convolutions introduces cumulative information loss, particularly severely affecting fine-grained details that small object detection depends upon. Targeting these limitations, we propose a novel feature aggregation module FIRC3, which leverages frequency domain transformations to enhance feature propagation and maintain the integrity of small object information throughout the detection pipeline, as illustrated in Fig. 5.

The input feature map X first generates intermediate representations through two parallel 1×1 convolutions. One branch undergoes cascaded FIRC (Frequency Iterative Refinement Convolution) transformation processing, while the other branch maintains the original feature flow. The module output is formulated as:

$$Y = W_3(\mathcal{M}(W_1 * X) + W_2 * X) \quad (13)$$

where W_1, W_2 represent channel projection operators, $\mathcal{M}(\cdot)$ denotes the transformation sequence composed of n cascaded FIRC operations, W_3 performs final channel mapping, $*$ denotes convolution operation, and the hidden channel dimension $C' = \lfloor e \cdot C_{out} \rfloor$ is controlled by expansion

Algorithm 3 FIRC3.

Require: Input feature $X \in \mathbb{R}^{C \times H \times W}$, iterations T

Ensure: Refined feature $Y \in \mathbb{R}^{C \times H \times W}$

```

1:  $X_1, X_2 \leftarrow \text{Split}(X)$  // Dual-path split
2:  $F \leftarrow X_1$ 
3: for  $t = 1$  to  $T$  do
4:   // Frequency-domain transformation
5:    $F_{\text{freq}} \leftarrow \text{FFT}(F)$ 
6:    $F_{\text{amp}} \leftarrow |F_{\text{freq}}|$  // Amplitude
7:    $F_{\text{phase}} \leftarrow \angle F_{\text{freq}}$  // Phase
8:   // Frequency-domain feature processing
9:    $F'_{\text{amp}} \leftarrow \text{Conv}_{1 \times 1}(F_{\text{amp}})$ 
10:   $F'_{\text{phase}} \leftarrow \text{Conv}_{1 \times 1}(F_{\text{phase}})$ 
11:  // Reconstruct and inverse transform
12:   $F'_{\text{freq}} \leftarrow F'_{\text{amp}} \cdot e^{i F'_{\text{phase}}}$ 
13:   $F' \leftarrow \text{IFFT}(F'_{\text{freq}})$ 
14:  // Residual connection
15:   $F \leftarrow F + \text{GELU}(\text{Conv}_{3 \times 3}(F'))$ 
16: end for
17:  $Y \leftarrow \text{Conv}_{1 \times 1}(\text{Concat}(F, X_2))$ 
18: return  $Y$ 

```

ratio e . The FIRC operator executes feature transformations in the frequency domain through a convex optimization framework. For a given feature map F , the FIRC operation first applies circular padding to enable seamless frequency domain processing, subsequently generating sparse and dense upsampled features through zero-interpolation upsampling $\mathcal{U}_s(F)$ and nearest-neighbor interpolation $\mathcal{I}_s(F)$ respectively. Define the frequency domain intermediate variable:

$$F_R = F[K] * \odot F[\mathcal{U}_s(F)] + F[\epsilon_b \cdot \mathcal{I}_s(F)] \quad (14)$$

where $F[\cdot]$ denotes the Fast Fourier Transform, K represents the depth-wise separable frequency domain convolution kernel initialized through softmax normalization, $(\cdot)^*$ denotes complex conjugate operation, $\odot$ represents element-wise product, $\epsilon_b = \sigma(b - 9.0) + \epsilon$ is the adaptive regularization term, $\sigma(\cdot)$ is the sigmoid function, and $\epsilon = 10^{-5}$ ensures numerical stability. This formula decomposes the original features into two pathways: the zero-interpolation path preserves precise information at original sampling positions, while the dense interpolation path provides smooth spatial continuity. Through the conjugate of the frequency domain convolution kernel and linear combination of both paths, adaptive fusion of multi-scale features is achieved. Based on F_R , the adaptive weight correction factor is computed:

$$W_{\text{inv}} = \frac{\text{Avg}_s(F[K] \odot F_R)}{\text{Avg}_s(|F[K]|^2) + \epsilon_b} \quad (15)$$

The averaging operator $\text{Avg}_s(\cdot)$ performs spatial downsampling on frequency domain responses, implementing frequency-selective aggregation by rearranging the spectrum into $s \times s$ blocks and averaging along the last dimension. The numerator term $F[K] \odot F_R$ computes the response intensity between the frequency domain convolution kernel and intermediate features, while the denominator term $|F[K]|^2$ represents

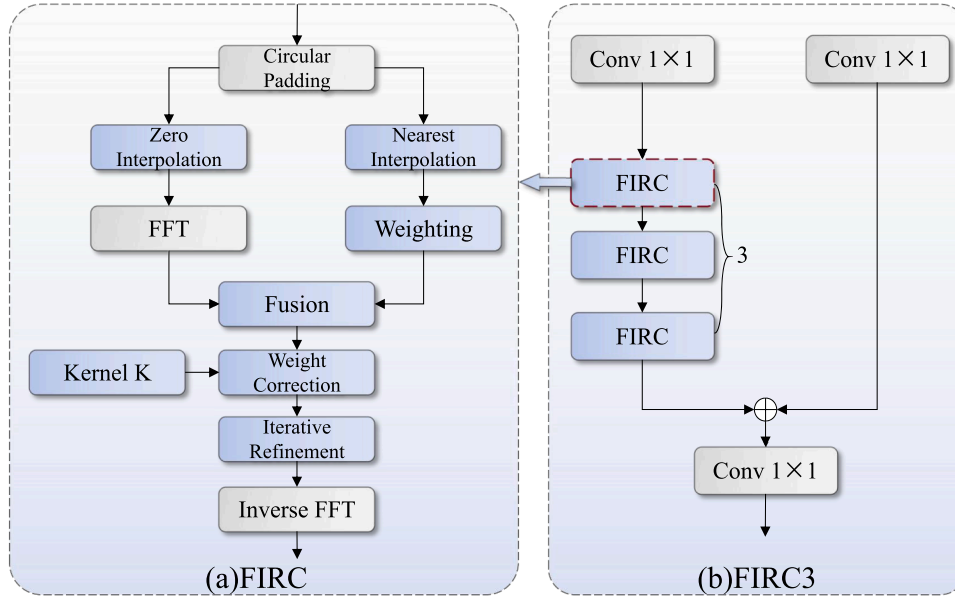


Fig. 5. FIRC3 block.

the squared magnitude of the frequency domain energy spectrum, serving a normalization role to prevent excessive amplification of frequency domain responses. The final output is obtained through frequency domain iterative solving:

$$\hat{F}_{\text{out}} = \mathcal{F}^{-1} \left[\frac{F_R - \mathcal{F}[K]^* \odot \text{Repeat}_s(W_{\text{inv}})}{\epsilon_b} \right] \quad (16)$$

where $\mathcal{F}^{-1}[\cdot]$ denotes the inverse Fourier transform, and $\text{Repeat}_s(\cdot)$ upsamples the downsampled weight factor to the original frequency domain resolution. The subtracted term $\mathcal{F}[K]^* \odot \text{Repeat}_s(W_{\text{inv}})$ in the numerator can be understood as refined correction of coarse-scale responses, implementing backpropagation through frequency domain conjugate operations, thereby suppressing low-frequency redundancy while maintaining high-frequency details. The entire transformation process essentially solves a frequency-domain-constrained least squares problem, adaptively balancing contributions of different frequency components and enabling the network to dynamically adjust sensitivity to high-frequency information of small objects. Periodization processing of the frequency domain convolution kernel is achieved through point spread function to optical transfer function conversion:

$$\mathcal{F}[K_{\text{circ}}] = \mathcal{F} \left[\text{Roll}(\text{ZeroPad}(K, (H_s, W_s)), (-\lfloor k/2 \rfloor, -\lfloor k/2 \rfloor)) \right] \quad (17)$$

where $\text{ZeroPad}(\cdot)$ pads the convolution kernel to the target feature map size, and $\text{Roll}(\cdot)$ performs circular shift to align the kernel center to the coordinate origin, thereby achieving true circular convolution. Frequency domain operations possess the natural advantage of global receptive fields compared to spatial domain convolutions, enabling the network to capture long-distance feature dependencies without increasing parameter count. Through learnable frequency domain convolution kernel parameterization, the module can dynamically adjust sensitivity to different frequency components, better adapting to feature distribution characteristics of objects at different scales.

Operating in the spectral domain grants FIRC3 an implicit global receptive field at $\mathcal{O}(N \log N)$ cost, while the iterative correction mechanism suppresses low-frequency redundancy and restores the boundary-defining high-frequency signal that repeated spatial convolutions progressively attenuate. The result is more precise localisation of small objects whose appearance is predominantly encoded in fine-scale edge structure.

4. Experiments

4.1. Experimental setup

All experiments were conducted on an NVIDIA GeForce RTX 3090 GPU. DFIR-DETR was implemented in PyTorch 2.2.2 / TorchVision 0.17.2, configured based on RT-DETR-R18. Training used AdamW with learning rate 1×10^{-4} and weight decay 5×10^{-4} , for 300 epochs with batch size 4, input resolution 640×640 , and mixed-precision training. Average GPU memory consumption was 11 GB and training required 7 h per dataset.

4.2. Benchmark datasets

These two benchmarks were not chosen arbitrarily. Prior work on small object detection tends to evaluate on a single domain, leaving open the question of whether reported gains reflect architectural improvements or dataset-specific overfitting. VisDrone and NEU-DET differ in sensor type, scene structure, object formation mechanism, and background statistics, but they both expose the same underlying bottleneck: insufficient high-frequency boundary representation for objects with limited pixel support. Demonstrating consistent gains across both settings provides stronger evidence that the proposed modules address a general architectural deficiency rather than a narrow application-specific one.

VisDrone (Zhu et al., 2022) contains 10,209 aerial images with 2.6M bounding boxes across 10 categories. The dataset splits into 6,471 training, 548 validation, and 3190 testing images, presenting challenges of large scale variations, dense occlusions, and viewpoint diversity, as illustrated in Fig. 6.

NEU-DET (Song & Yan, 2013) comprises 1800 grayscale images of hot-rolled steel surfaces with six defect types (crazing, inclusion, patches, pitted surface, rolled-in scale, scratches), each with 300 images at 200×200 pixels. Challenges include subtle texture variations, complex backgrounds, and high inter-class similarity, as illustrated in Fig. 7.

4.3. Ablation study

Table 1 reports ablation results on NEU-DET under two complementary protocols: sequential module addition and single-module removal

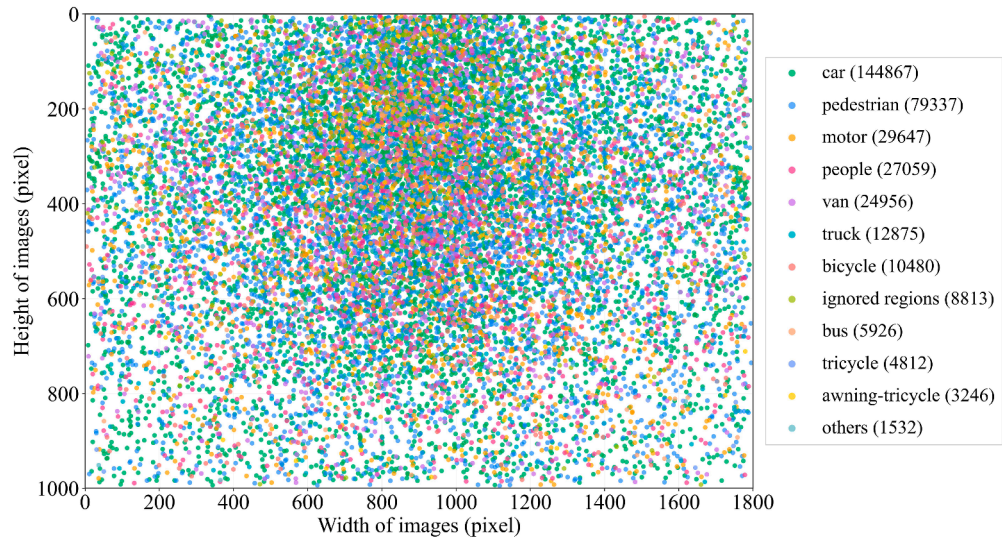


Fig. 6. VisDrone object instance spatial distribution.

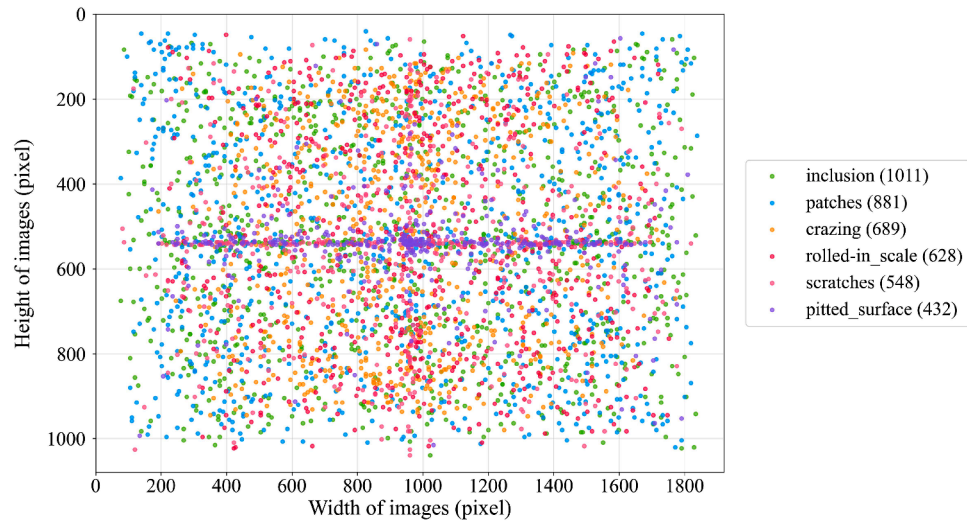


Fig. 7. NEU-DET defect instance spatial distribution.

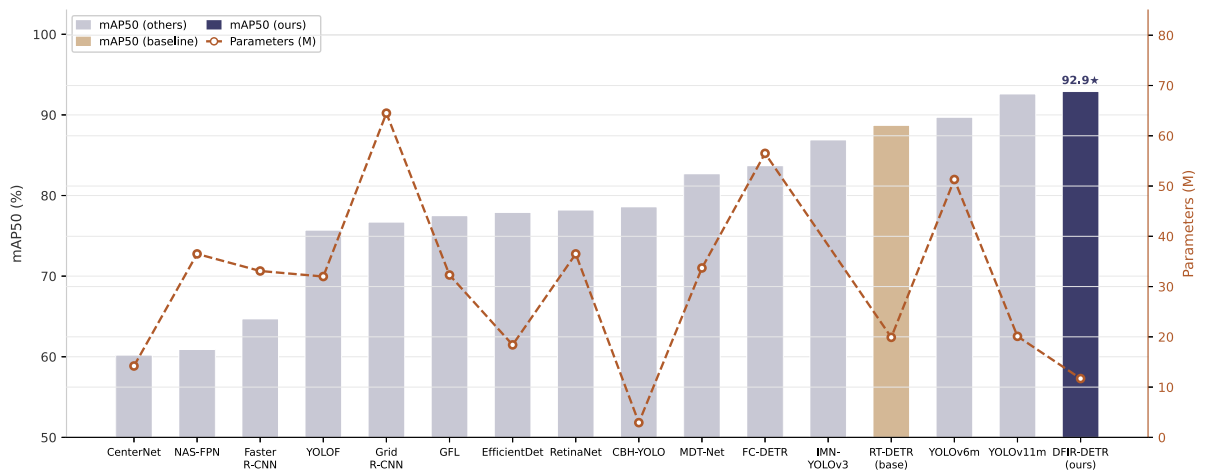


Fig. 8. Comparison on NEU-DET.

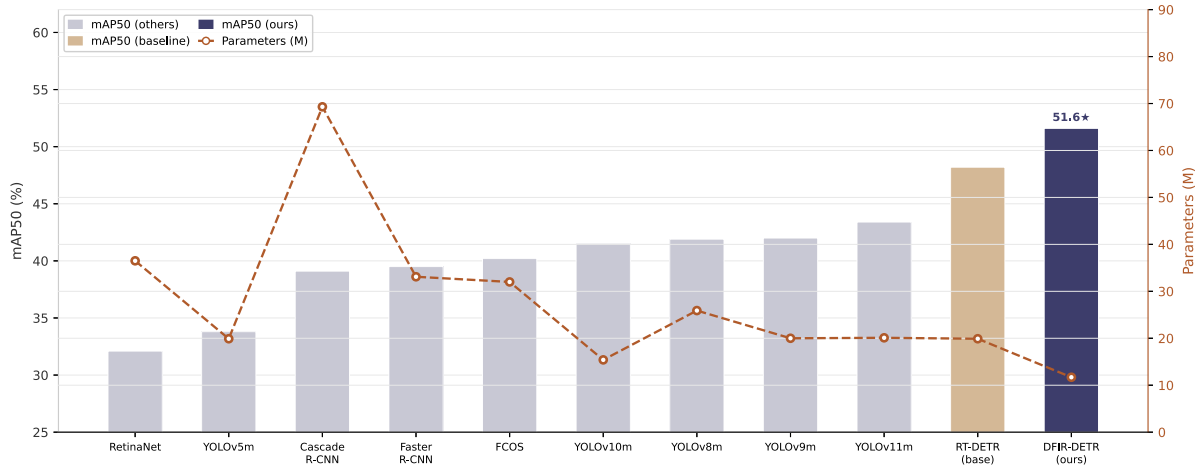


Fig. 9. Comparison on VisDrone.

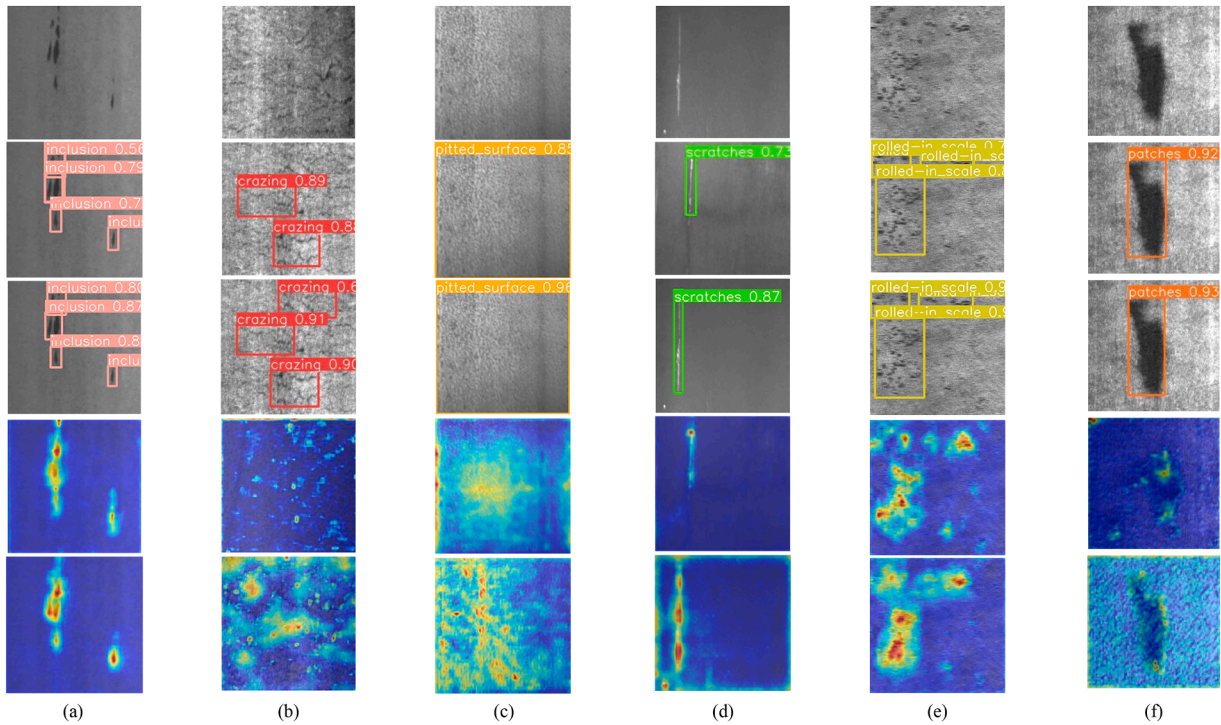


Fig. 10. Qualitative visualization comparison on NEU-DET dataset.

from the complete model. DCFA alone lifts mAP50 by 1.4 points over the RT-DETR baseline while trimming parameters from 19.9M to 13.7M; the 31.2% reduction comes from the sparse attention structure, not from any sacrifice in representational depth. Substituting only DFPPN into the baseline reaches 90.9% mAP50, with the Ri category gaining 11.2 points over baseline, the largest single-category swing in the table; this is consistent with ANUP counteracting the norm inflation that standard up-sampling introduces at that scale. FIRC3 alone reaches 90.5% mAP50 at the lowest GFLOPs of any single-module configuration, reflecting the computational advantage of spectral operations over repeated spatial convolutions. The complete model reaches 92.9% mAP50 and 65.9% mAP50:95; the 7.7-point mAP50:95 margin over baseline outpaces the mAP50 margin, which points to boundary localisation rather than coarse classification as the primary beneficiary. Parameter count and GFLOPs land at 11.7M and 47.2, down 41.2% and 17.2% from the starting point.

Removal experiments invert the construction protocol. Taking DCFA out of the full model costs 1.8 mAP50 points; removing DFPPN costs 1.2

points; removing FIRC3 costs 0.5 points. The ordering matches the sequential results: DCFA contributes most on NEU-DET, where content-adaptive sparsification has the sharpest effect on defect boundary discrimination, and confirms that the gains are not an artefact of module interaction alone.

4.4. Comparative performance analysis and qualitative visualisation

On NEU-DET, DFIR-DETR reaches 92.9% mAP50, surpassing the RT-DETR baseline by 4.2 points and YOLOv11m by 0.3 points while reducing parameters to 11.7M, a 77.2% reduction relative to YOLOv6m at substantially higher accuracy, as shown in Table 4. The mAP50:95 gap over baseline is 7.7 points, larger than the mAP50 gap, indicating that frequency-domain boundary recovery contributes more to tight localisation than to coarse detection. Beyond general-purpose detectors, Table 4 also incorporates three methods developed specifically for steel surface inspection: IMN-YOLOv3 (Chen et al., 2022), FC-DETR (Li et al., 2025),

Table 2

Backbone comparison on VisDrone dataset.

Model	mAP50	Ped	Peo	Bic	Car	Van	Tru	Tri	Awn	Bus	Mot	mAP95	Params	GFLOPs	FPS
Strip (Guo et al., 2023b)	48.4	53.8	36.6	23.9	81.4	54.5	57.7	35.7	23.9	68.8	49.9	28.8	14.3	47.7	36.1
MambaOut (Yu & Wang, 2024)	46.7	51.1	34.0	22.9	80.2	51.8	53.9	34.4	24.4	65.9	48.2	27.8	15.9	41.9	34.7
GlobalFilter (Rao et al., 2022)	48.4	53.6	36.7	25.5	81.4	53.4	55.7	34.1	27.4	66.3	50.0	28.7	16.9	46.5	36.9
AP (Dai et al., 2017)	48.6	53.9	35.8	23.8	81.8	54.6	56.6	34.8	28.1	67.4	49.8	28.9	13.3	47.0	36.9
FDT (Mao et al., 2023)	47.2	51.7	34.5	24.0	81.5	53.4	53.8	33.6	25.2	65.8	48.1	28.4	15.4	52.5	32.3
DTAB (Liu et al., 2023b)	48.8	55.1	37.9	27.0	82.2	53.6	56.3	34.4	24.7	66.1	51.0	29.3	18.8	57.3	59.9
CAMixer (Tu et al., 2023)	49.1	53.9	36.8	25.1	81.8	53.5	58.4	35.0	26.8	69.2	50.1	29.4	13.2	43.3	43.2
DCFA (ours)	49.4	55.4	38.1	27.6	82.2	55.1	56.4	35.3	24.4	68.0	51.9	30.1	13.7	51.7	34.2

Table 3

Backbone comparison on NEU-DET dataset.

Model	mAP50	Cr	In	Pa	Ps	Ri	Sr	mAP50:95	Params	GFLOPs	FPS
EfficientViM (Zhu et al., 2024)	87.1	59.3	93.1	96.2	92.8	83.2	98.4	57.5	14.5	47.9	73.1
MambaOut (Yu & Wang, 2024)	82.3	64.2	89.4	96.0	93.1	75.5	94.8	54.0	15.9	41.8	59.0
GlobalFilter (Rao et al., 2022)	88.0	67.2	91.0	96.2	95.5	82.2	95.8	58.5	16.8	46.5	72.3
ELGCA (Song et al., 2024)	88.2	62.0	94.6	94.8	92.1	87.3	98.2	60.5	13.9	46.9	38.2
FDT (Mao et al., 2023)	84.5	57.4	91.4	97.1	92.7	74.0	94.3	55.3	15.4	52.5	32.7
DTAB (Liu et al., 2023b)	84.5	53.5	92.1	95.7	93.1	75.4	96.9	54.3	18.8	57.3	31.8
HDRAB (Wang et al., 2023c)	84.8	49.0	93.6	96.1	90.1	82.4	97.5	56.0	15.6	53.2	37.6
MSN (Huang et al., 2023)	86.0	53.6	93.2	95.8	94.1	81.2	98.3	59.2	16.6	49.3	32.4
FCA (Qin et al., 2020)	87.4	63.8	93.4	94.8	91.9	83.9	96.8	58.5	15.4	52.6	31.2
RAB (Yang et al., 2021)	83.6	48.7	93.1	95.5	90.4	77.8	95.9	54.8	13.3	53.2	34.9
DCFA (ours)	90.1	78.9	91.2	97.0	96.8	84.0	98.6	60.8	13.7	51.7	37.0

Table 4

Comprehensive comparison on NEU-DET dataset.

Model	mAP50	Cr	In	Pa	Ps	Ri	Sr	mAP50:95	Params	GFLOPs	FPS
YOLOv6m (Li et al., 2023a)	89.7	72.0	88.2	96.8	95.2	89.1	96.9	62.1	51.3	158.3	80.4
YOLOv11m (Ultralytics, 2024)	92.6	84.3	90.3	97.0	94.4	92.0	97.1	72.4	20.1	68.0	79.6
YOLOF (Chen et al., 2021)	75.7	39.7	83.0	90.2	86.1	58.2	97.1	43.7	32.0	52.6	80.5
Faster R-CNN (Ren et al., 2017b)	64.7	30.7	66.9	91.3	67.0	53.9	78.2	29.9	33.1	208.1	18.4
RetinaNet (Lin et al., 2017b)	78.2	50.4	77.8	93.3	89.5	63.9	94.5	43.3	36.5	210.0	68.2
Grid R-CNN (Lu et al., 2019)	76.7	39.7	82.8	92.5	87.4	60.2	97.5	44.4	64.5	270.2	30.5
GFL (Li et al., 2020)	77.5	37.4	84.1	93.9	90.9	62.8	95.9	44.5	32.3	195.2	45.8
NAS-FPN (Ghiasi et al., 2019)	60.9	29.2	50.0	82.8	79.3	55.6	68.4	26.0	36.5	84.5	–
EfficientDet (Tan et al., 2020)	77.9	49.8	78.9	93.6	86.8	65.8	92.2	44.6	18.4	55.1	48.9
CenterNet (Zhou et al., 2019)	60.2	22.2	59.3	77.8	77.1	29.7	95.2	27.4	14.2	38.3	48.3
IMN-YOLOv3 (Chen et al., 2022)	86.9	72.0	86.9	94.8	94.3	80.6	93.2	–	–	–	–
FC-DETR (Li et al., 2025)	83.7	67.2	85.8	91.9	88.2	72.5	96.6	–	56.5	143	–
MDT-Net (Wang et al., 2025)	82.7	50.2	87.4	92.7	86.0	83.0	96.9	55.0	33.7	26.6	18
CBH-YOLO (Gao et al., 2025)	78.6	53.2	83.0	93.4	82.5	64.6	–	45.8	2.9	7.4	137.1
RT-DETR (base)	88.7	78.4	87.0	96.0	94.6	78.0	98.1	58.2	19.9	57.0	36.9
DFIR-DETR (ours)	92.9	84.0	93.7	96.8	98.4	84.9	99.3	65.9	11.7	47.2	38.0

Table 5

Neck/CCFF module comparison on NEU-DET dataset.

Model	mAP50	Cr	In	Pa	Ps	Ri	Sr	mAP50:95	Params	GFLOPs	FPS
ASF (Yang et al., 2022a)	85.8	56.8	94.7	95.5	93.2	76.6	97.8	57.8	20.2	61.5	39.1
SDI (Zhao et al., 2023)	85.2	54.2	92.7	96.0	93.6	77.2	97.2	57.6	19.9	57.1	35.7
Goldyolo (Wang et al., 2023a)	87.4	62.3	94.6	95.0	94.3	80.6	97.5	58.1	22.3	60.0	32.7
HSFPN (Li et al., 2023d)	85.2	55.8	92.7	94.9	93.1	78.7	96.2	55.9	18.1	53.3	41.0
CGAFusion (Guo et al., 2023a)	87.0	58.7	93.8	95.6	93.9	83.9	96.4	59.4	20.4	59.2	38.9
PSFM (Sun et al., 2023)	85.9	55.9	94.9	95.1	93.2	79.3	97.3	58.7	22.4	71.2	26.9
GLSA (Zhang et al., 2023b)	85.1	49.3	93.8	95.9	92.7	81.1	97.9	58.2	22.0	63.7	31.1
CTrans (Yan et al., 2023)	81.6	50.8	87.6	93.3	93.2	71.9	92.7	51.8	29.5	57.4	86.9
MAFFN (Liu et al., 2023a)	83.6	52.6	92.6	95.2	90.4	74.8	96.1	55.5	22.9	56.3	33.8
MSGa (Wang et al., 2023b)	87.8	66.1	93.1	95.3	93.5	82.5	96.6	58.9	22.5	71.1	36.9
FSA (Li et al., 2023c)	85.0	56.6	94.2	96.2	92.9	73.2	97.1	57.4	22.6	57.0	38.6
MFM (Hu et al., 2023)	85.8	57.1	93.7	95.8	93.8	77.0	97.3	56.5	19.7	55.7	37.0
DFPN (ours)	90.9	72.1	94.5	97.0	94.4	89.2	98.3	62.6	19.7	56.7	33.7

Table 6

RepC3 module comparison on NEU-DET dataset.

Model	mAP50	Cr	In	Pa	Ps	Ri	Sr	mAP50:95	Params	GFLOPs	FPS
ConvXCC3 (Ding et al., 2021)	83.5	51.0	93.3	94.9	93.4	70.3	98.3	55.4	19.9	57.0	41.5
DBBC3 (Han et al., 2023)	87.2	62.9	91.6	94.6	91.7	85.3	97.3	58.0	19.9	57.0	47.0
DGCST (Chen et al., 2023)	87.8	65.2	90.5	95.4	95.3	77.6	98.4	56.7	18.5	50.2	42.1
LITC3 (Liu et al., 2023c)	84.3	45.7	94.4	95.6	93.9	78.9	97.5	53.3	19.8	56.0	39.8
FIRC3 (ours)	90.5	76.2	94.6	95.8	92.8	84.7	98.6	63.2	18.2	48.4	47.8

Table 7
Comprehensive comparison on VisDrone dataset.

Model	mAP50	Ped	Peo	Bic	Car	Van	Tru	Tri	Awn	Bus	Mot	mAP50:95	Params	GFLOPs	FPS
Faster R-CNN (Ren et al., 2017b)	39.5	39.2	33.0	20.8	74.1	45.8	39.0	30.2	15.1	55.3	42.6	25.6	33.1	208.1	16.5
RetinaNet (Lin et al., 2017b)	32.1	32.1	25.2	13.0	71.4	37.8	35.9	21.5	9.7	43.3	31.9	20.6	36.5	210.0	41.3
FCOS (Tian et al., 2019)	40.2	45.6	34.6	19.4	78.3	44.2	38.9	28.2	12.9	52.5	47.8	26.1	32.0	203.8	–
Cascade R-CNN (Cai & Vasconcelos, 2018)	39.1	39.1	31.3	19.3	74.2	46.3	39.2	30.8	13.8	55.1	42.0	26.0	69.3	241.7	13.3
YOLOv5m (Ultralytics, 2021)	33.8	42.6	33.5	11.5	74.3	34.8	30.5	18.4	10.9	40.8	40.9	18.9	19.9	64.2	60.0
YOLOv8m (Jocher et al., 2023)	41.9	44.7	34.7	15.9	80.8	45.9	39.0	31.6	16.2	63.1	47.5	25.1	25.9	78.9	56.3
YOLOv9m (Wang & Liao, 2024)	42.0	17.9	37.1	17.8	82.1	50.6	43.6	37.4	20.8	62.7	49.8	27.6	20.0	76.8	62.2
YOLOv10m (Wang et al., 2024)	41.5	43.6	35.0	15.4	80.5	48.0	40.1	29.9	18.1	58.3	46.6	25.3	15.4	59.1	65.7
YOLOv11m (Ultralytics, 2024)	43.4	48.0	36.2	18.2	82.0	48.6	40.0	32.8	19.3	60.4	49.3	26.5	20.1	68.0	64.5
RT-DETR (base)	48.2	53.2	37.3	25.1	81.3	53.9	56.8	31.8	24.7	67.4	50.1	29.1	19.9	57.0	48.1
DFIR-DETR (ours)	51.6	56.6	39.9	25.1	83.6	55.4	57.8	35.6	36.4	69.6	55.6	31.6	11.7	47.2	40.1

Table 8
Comparison with UAV-specific and small-object-oriented methods on VisDrone dataset (640 × 640 input).

Model	mAP50	mAP50:95	Params	GFLOPs	FPS
MFP-YOLO (Li et al., 2023b)	46.0	–	1.48	–	28.5
CRL-YOLOv5 (Zhang et al., 2024)	39.2	22.3	–	–	27
AUP-DETR (Xu et al., 2025)	48.5	29.9	22.7	83.6	–
WDFS-DETR (Liu & Xie, 2025)	47.5	–	19.9	53.7	–
TCF-DETR (Lei et al., 2025)	49.9	29.9	19.8	61.9	–
RT-DETR (base)	48.2	29.1	19.9	57.0	48.1
DFIR-DETR (ours)	51.6	31.6	11.7	47.2	40.1

and MDT-Net (Wang et al., 2025). Despite their domain-specific design, the best-performing specialist method, IMN-YOLOv3, achieves 86.9% mAP50, leaving a gap of at least 6.0 points relative to DFIR-DETR's 92.9%. Concurrently, FC-DETR requires 56.5M parameters to achieve 83.7%, nearly five times the footprint of our model at comparable or lower accuracy. The pattern suggests that frequency-domain boundary modelling addresses a representational gap that domain-specific architectural choices alone have not closed.

At the component level, Table 3 shows that DCFA achieves 90.1% mAP50 on NEU-DET, outperforming ELGCA by 1.9 points despite comparable parameter counts, with particularly strong gains on crazing (+16.9 points over ELGCA). On VisDrone, Table 2 shows that DCFA reaches the highest mAP50:95 of 30.1% among all backbone alternatives, suggesting that content-adaptive attention allocation provides more consistent localisation across the ten aerial categories than geometry-driven or token-reduction approaches. As reported in Table 5, DFPN achieves 90.9% mAP50 on NEU-DET, a 3.1-point margin over MSGA, with the Ri category showing the largest per-class gain (+6.7 points), consistent with ANUP preserving the fine-scale edge structure that rolled-in scale defects require. Table 6 shows that FIRC3 reaches 90.5% mAP50, outperforming DGCST by 2.7 points, while also achieving the lowest GFLOPs among all compared fusion modules, as illustrated in Fig. 8.

On VisDrone, DFIR-DETR achieves 51.6% mAP50, a 3.4-point improvement over the RT-DETR baseline and 8.2 points over YOLOv11m, as reported in Table 7. The awning-tricycle category shows the largest per-class gain (+11.7 points over baseline), a category combining small physical size with an unusual aspect ratio where receptive-field adaptation and high-frequency boundary retention matter most, as illustrated in Fig. 9. Table 8 further compares DFIR-DETR against UAV-specific detectors under the identical 640 × 640 input protocol. DFIR-DETR surpasses MFP-YOLO and CRL-YOLOv5 by 5.6 and 12.4 percentage points in mAP50 respectively, while achieving the lowest parameter count among all compared Transformer-based methods.

Fig. 10 presents qualitative detection results and GradCAM activation maps on NEU-DET across all six defect categories. Each column corresponds to one category: (a) inclusion, (b) crazing, (c) pitted surface, (d) scratches, (e) rolled-in scale, and (f) patches. For each column, the top row shows the ground truth annotation, the second and third rows

show detection results from the RT-DETR baseline and DFIR-DETR respectively, and the bottom two rows show the corresponding GradCAM activation maps.

Across all six categories, DFIR-DETR produces tighter bounding boxes at higher confidence than the baseline. For spatially elongated defects such as scratches (d) and rolled-in scale (e), the baseline tends to produce loose or fragmented boxes, whereas DFIR-DETR recovers more precise boundary extents. The GradCAM maps tell the same story from the feature side: baseline activations spread broadly across the feature map, whereas DFIR-DETR pulls high-activation regions onto defect boundaries. This contrast is most pronounced in the inclusion (a) and pitted surface (c) categories, where subtle low-contrast textures demand precise high frequency boundary discrimination. The tighter activation footprint is a direct visual signature of the spectral boundary reinforcement that FIRC3 introduces.

5. Conclusion

Small object detection in complex, cross-scene environments exposes fundamental tensions in neural network design that cannot be resolved by scaling existing architectures. This work argues that three specific structural properties (uniform attention allocation, amplitude drift during multi-scale fusion, and spectral attenuation through repeated spatial convolution) collectively account for a substantial portion of the performance gap between current detectors and the demands of real-world deployment. DFIR-DETR addresses each through principled architectural interventions: content-adaptive sparse attention in DCFA, norm-preserving feature pyramid construction in DFPN, and frequency-domain iterative refinement in FIRC3. The resulting framework demonstrates that targeted, theoretically motivated modifications to backbone, neck, and fusion modules can simultaneously improve accuracy, reduce parameters, and maintain real-time throughput without relying on larger training data or heavier model capacity.

Three concrete directions emerge from the limitations of the current design. The frequency-domain computation in FIRC3 stops at the encoder: decoder cross-attention operates entirely in the spatial domain, and there is no mechanism to bias query-key interactions toward boundary-informative spectral components. Introducing a spectral regularisation term into decoder attention could tighten localisation for the sub-16-pixel targets where spatial cues are weakest. The norm-preserving argument behind DFPN currently assumes nearest-neighbour upsampling; the derivation does not carry over to learnable operators such as CARAFE or DySample, whose sampling grids introduce non-uniform amplitude effects that the present normalisation factor does not account for. Finally, DCFA commits to a single sparsity ratio for an entire feature map, which treats a defect cluster and an empty background patch as equally informative. At a modest spatial resolution, Patch-level ratio prediction could redistribute attention budget more precisely without the overhead of full dense computation.

CRedit authorship contribution statement

Bo Gao: Writing – original draft, Methodology, Investigation, Formal analysis; **Jingcheng Tong:** Visualization, Software, Methodology, Investigation, Formal analysis; **Kingsheng Chen:** Writing – review & editing, Supervision; **Han Yu:** Writing – review & editing, Supervision; **Zichen Li:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge support from the [National Natural Science Foundation of China](#) (Grant No. 62472040), the Open Project Funding of Key Laboratory of Mobile Application Innovation and Governance Technology, Ministry of Industry and Information Technology (Grant No. 2024IFS080607-A), the Beijing Institute of Graphic Communication High-level Talent Program (Grant No. 27170024003), the Data Intelligence and Security Innovation Team Program of Inner Mongolia Autonomous Region Higher Education (Grant No. NMGIRT2506), the Beijing Natural Science Foundation Joint Fund (Grant No. L251065), the General Science and Technology Project of the [Beijing Municipal Education Commission](#) (Grant No. KM202410015001), the Copyright Research Project of the China Copyright Protection Center (Grant No. BQ2024017), and the 2025 Beijing - Hong Kong - Macao University Exchange Program (Grant No. 11000025T000003319739), and the Key Project of the National Social Science Foundation of China (2025), Research on Agricultural Product Supply Chain Security of the China-Mongolia-Russia Economic Corridor (Grant No. 25AGJ021).

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